Mathematics: analysis and approaches
Standard level
Paper 1
4.1–4.6 Statistics and probability

13 October 2026

W4ss Academy practice set 2 | **Topic** 4.1–4.6 Statistics and probability | **Ref.** W4SS–MAA–T4–P1–02

Student name

30 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- You are **not** permitted access to any calculator for this paper.
- This paper has a Section A only. Answer all questions. Answers must be written within the answer boxes provided.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[26 marks]**.

Online revision: waslyl.github.io/Test-Question-banks
W4ss Academy
Practice papers

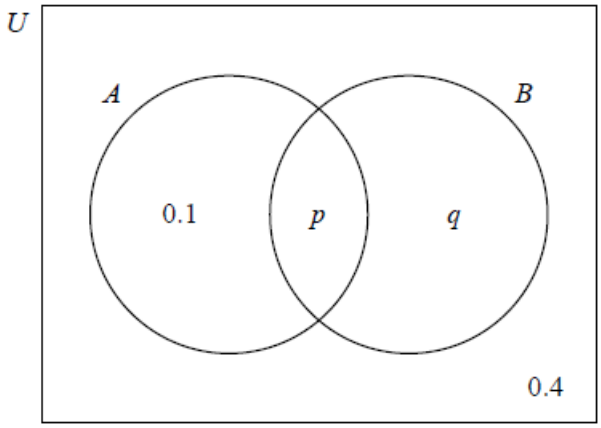
Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 6]

The following Venn diagram shows the events A and B, where $P(A) = 0.3$. The values shown are probabilities.



(a) Find the value of p. [2]

(b) Find the value of q. [2]

(c) Find $P(A' \cup B)$. [2]

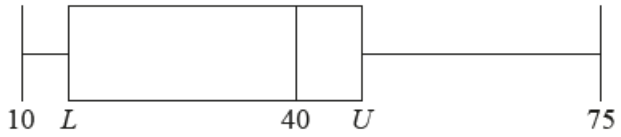
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(Question 1 continued)

2. [Maximum mark: 5]

A research student weighed lizard eggs in grams and recorded the results. The following box and whisker diagram shows a summary of the results where L and U are the lower and upper quartiles respectively.

diagram not to scale



The interquartile range is 20 grams and there are no outliers in the results.

(a) Find the minimum possible value of U. [3]

(b) Hence, find the minimum possible value of L. [2]

3. [Maximum mark: 6]

Events A and B are such that $P(A) = 0.3$ and $P(B) = 0.8$.

(a) Determine the value of $P(A \cap B)$ in the case where the events A and B are independent. [1]

(b) Determine the minimum possible value of $P(A \cap B)$. [3]

(c) Determine the maximum possible value of $P(A \cap B)$, justifying your answer. [2]

4. [Maximum mark: 9]

A set of data comprises of five numbers x_1, x_2, x_3, x_4, x_5 which have been placed in ascending order.

(a) Recalling definitions, such as the Lower Quartile is the $\frac{n+1}{4}th$ piece of data with the data placed in order, find an expression for the Interquartile Range. [2]

(b) Hence, show that a data set with only 5 numbers in it cannot have any outliers. [5]

(c) Give an example of a set of data with 7 numbers in it that does have an outlier, justify this fact by stating the Interquartile Range. [2]

(This question continues on the following page)

(Question 4 continued)

Sources of the questions

This practice paper was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Mathematics: analysis and approaches Questionbank (standard level, topic 4 Statistics and probability) and cover subtopics 4.1 to 4.6. They remain the property of the International Baccalaureate Organization and must not be published or sold.

Section A — question 1	19M.1.SL.TZ1.S_1 (former Mathematics SL course) · 4.6 Venn diagram, combined events
Section A — question 2	21M.1.SL.TZ1.4 · 4.2 box and whisker diagram, outliers
Section A — question 3	22N.1.SL.TZ0.6 · 4.6 independent events, combined events
Section A — question 4	EXM.1.SL.TZ0.2 · 4.2, 4.3 quartiles, interquartile range, outliers